EECS 298 - Quantum Info & Architecture

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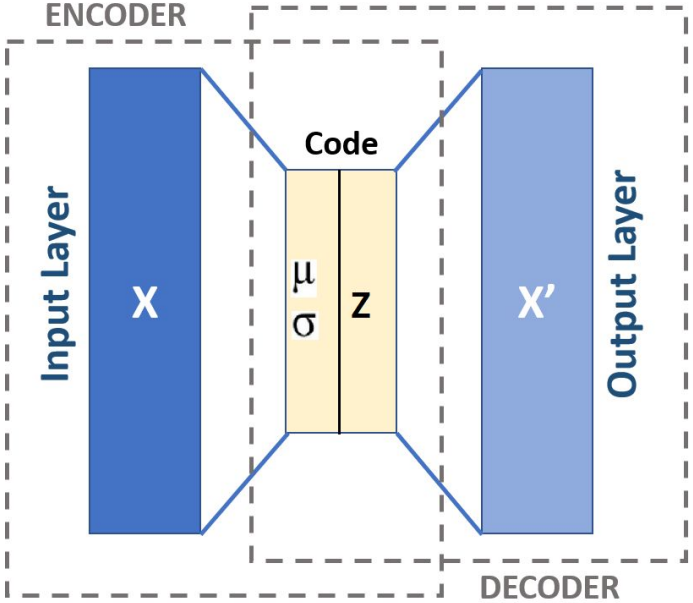
Quantum Error Correction: CSS Codes and QSCD in Qiskit

Introduction

- Qubit entanglement is conserved for each qubit (**No-cloning theorem**)
  - Preparation, conditional gates, noise and measurement are entanglement
- Entanglement provides quantum advantage
  - Encoding space from n separable qubits (n bits) to n fully entangled qubits (2^n bits)
  - Noise is catastrophic:
    - Full entanglement to m environment qubits = 2^m - fold reduction in capacity

Application motivation - Quantum VAE

- Encode to a **small** latent space, transform, decompress to results
- **Quantum VAE:**
  - **Classical** encoder-decoder
  - **Quantum gates** in latent space
    - Smaller more uniform qubits
    - Shorter gate depth
    - Better fidelity
  - Encoder-decoder tailored to syndromes of the quantum circuit

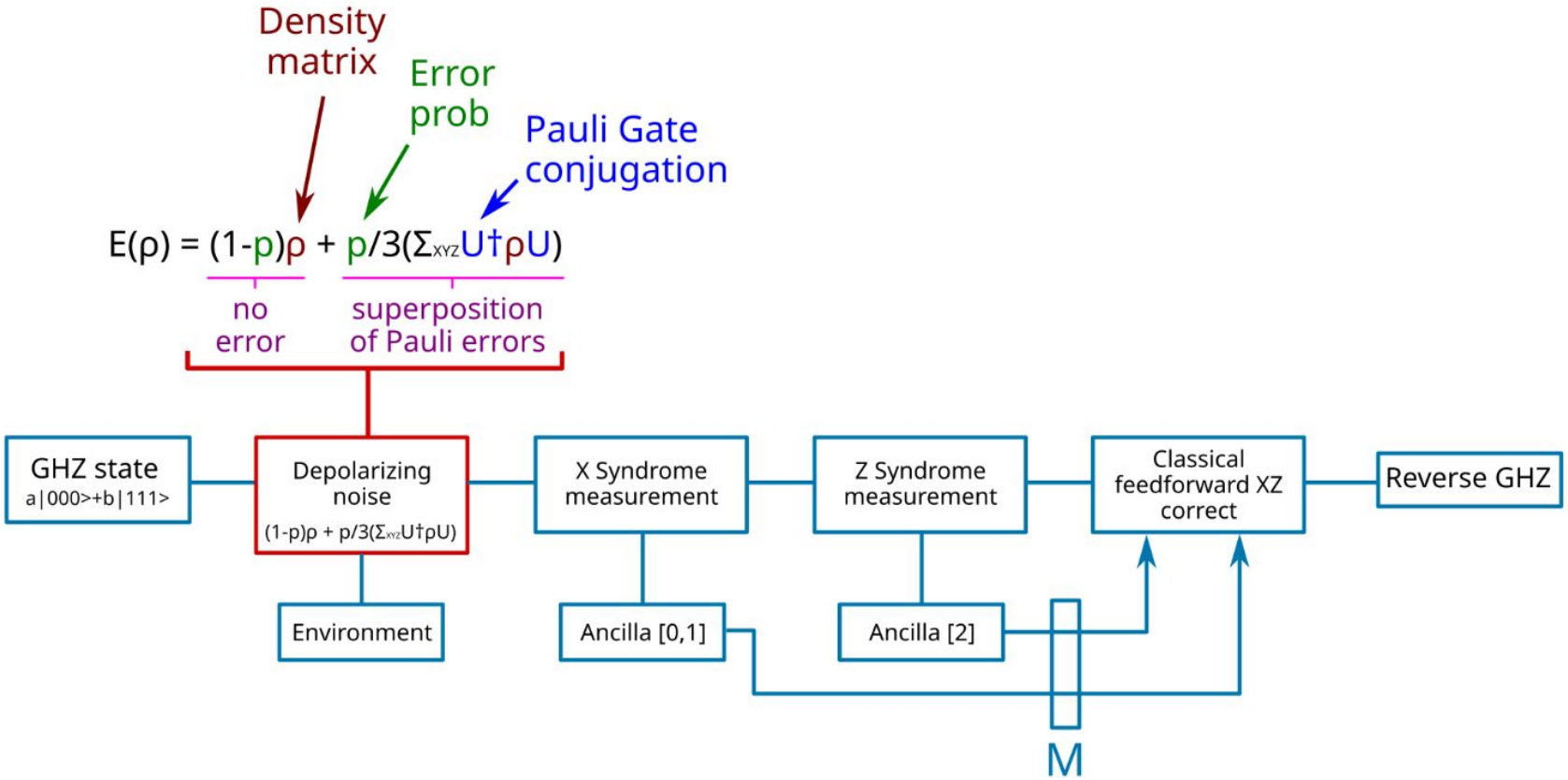


Implementation

- **CSS (Calderbank-Shor-Steane)** codes protect quantum information by spreading one logical qubit across multiple physical qubits
- CSS codes elegantly handle X and Z errors independently.
- **Quantum Successive Cancellation Decoding (QSCD)** identifies and fixes errors one at a time without ever directly reading the data.
  - Classical feedforward correction
    - Proven optimal: Renes 2012*
  - Successive: corrections applied sequentially
- With CSS and QSCD, we can recover the original qubit with high fidelity as long as the noise stays below a certain threshold.

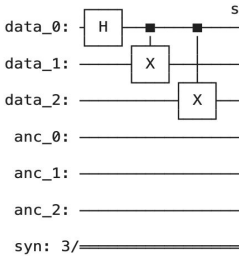
*Joseph M. Renes and Mark M. Wilde. Polar codes for private and quantum communication over arbitrary channels.

Theoretical Explanation

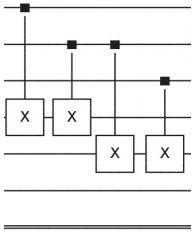


CSS Codes

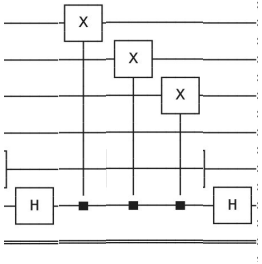
How It Works



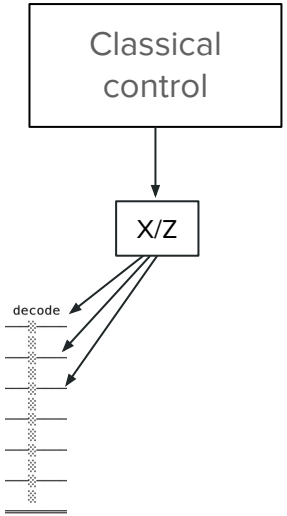
GHZ state



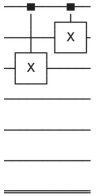
X syndrome
Parity of qubit pairs



Z syndrome
Phase kickback
detect (OR circuit ⅓)



Classical
feedforward



GHZ state
reversal

Error Correction

Ancilla bits

(s0, s1, sz)

Syndrome (s0,s1)	Error	Correction
(1, 0)	X on qubit 0	X on qubit 1
(1, 1)	X on qubit 1	X on qubit 0
(0, 1)	X on qubit 2	X on qubit 2

Syndrome (sz)	Error	Correction
1	Z on any qubit	Z on qubit 0

Y = iXZ

Syndrome (s0,s1,sz)	Error	Correction
(1, 0, 1)	Y on qubit 0	X q1 + Z q0
(1, 1, 1)	Y on qubit 1	X q0 + Z q0
(0, 1, 1)	Y on qubit 2	X q2 + Z q0

Qiskit Implementation

```
# CSS Encoder
qc.h(data[0])              # |+⟩
qc.cx(data[0], data[1])    # CNOT
qc.cx(data[0], data[2])    # CNOT

# Z-type stabilizers
qc.cx(data[0], anc[0])
qc.cx(data[1], anc[0])     # Z₀Z₁
qc.cx(data[1], anc[1])
qc.cx(data[2], anc[1])     # Z₁Z₂

# X-type stabilizer
qc.h(anc[2])
qc.cx(anc[2], data[0])
qc.cx(anc[2], data[1])
qc.cx(anc[2], data[2])     # X₀X₁X₂
qc.h(anc[2])

# Measure syndrome
qc.measure(anc[0], syn[0])
qc.measure(anc[1], syn[1])
qc.measure(anc[2], syn[2])
```

Design Decisions

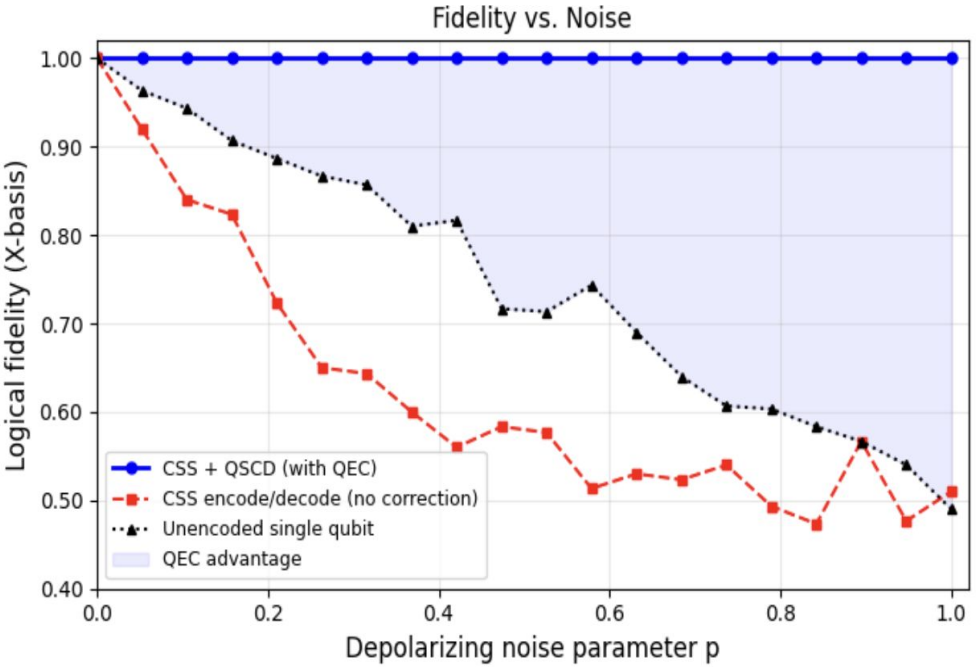
Density matrix sim

Needed for mixed states under depolarizing noise. Handles mixed states after noise; state vector would give wrong answers

Two-pass approach

Mid-circuit classical feedforward implemented by saving full post-measurement density matrix and resuming in Pass 2.

Results & Discussion



- ✓ All 9 single-qubit Pauli errors (X/Y/Z × 3 qubits) corrected with fidelity = 1.000
- ✓ p = 0 baseline: fidelity = 1.0 confirms no pipeline bugs
- → Noise on ancilla qubits corrupts syndrome: data-only noise model is critical

Results & Discussion

Error	Qubit	Syn(s0,s1,sz)	X fix	Z fix	Result
X	0	100	('X', 1)	None	✓
X	1	110	('X', 0)	None	✓
X	2	010	('X', 2)	None	✓
Z	0	001	None	('Z', 0)	✓
Z	1	001	None	('Z', 0)	✓
Z	2	001	None	('Z', 0)	✓
Y	0	101	('X', 1)	('Z', 0)	✓
Y	1	111	('X', 0)	('Z', 0)	✓
Y	2	011	('X', 2)	('Z', 0)	✓

All single-qubit errors corrected: ✓ YES

Table confirms all 9 single-qubit Pauli errors corrected with a checkmark

Summary

- CSS encoder: 2 CNOTs fan 1 logical qubit into a 3-qubit GHZ state
- Two-pass density matrix simulation faithfully implements QSCD without dynamic circuits
- Data-only noise is essential — ancilla noise corrupts syndrome readout

Thank You :)

Questions?